Lab - Configure Security Features in Windows and Linux

Objectives

Part 1: Update Windows and Linux

Part 2: Windows Local Security Policy

Part 3: Configure Firewall Rules

Part 4: Install and Run Applications

Required Resources

1 Windows 10 PC

This CSE-LABVM virtual machine

Internet access

Background / Scenario

In this lab, you will update both Windows and Linux systems. You will configure local security policy and configure firewall rules in Windows. In Linux, you will install two applications: chkrootkit and lynis.

Instructions

Part 1: Update Windows and Linux

New flaws and vulnerabilities are discovered all the time. It is a good idea to keep your PC up to date to mitigate the exploitation of the known vulnerabilities.

Step 1: Verify connectivity between the CSE-LABVM and the Windows host computer.

In this step, you will verify connectivity to the internet so that you can download updates. Furthermore, you will verify connectivity between **CSE-LABVM** and Windows PC so that you can perform tasks later in this lab.

a. Before launching the **CSE-LABVM**, select it and then choose **Settings** > **Network**. For **Adapter 1**, change the **Attached to:** option to **Bridged Adapter**. You can then choose the adapter. Many computers have two adapters: one for wireless networks and the other for wired networks. Choose the one your computer is using to connect to the internet.

b. Launch the CSE-LABVM and wait for it to boot.

c. On the CSE-LABVM, open a terminal and enter **ip address** to determine its IP address.

d. On your Windows host computer, open a command prompt and enter **ipconfig** to determine its IP address.

Record the IP address for both CSE-LABVM and Windows PC.

CSE-LABVM:

Windows PC:

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e. From the respective command prompts, ping a website of your choice to verify that the Windows host and **CSE-LABVM** can reach the internet.

f. Verify that the Windows host can ping the **CSE-LABVM**.

g. From **CSE-LABVM**, attempt to ping the Windows host. The **CSE-LABVM** may not be able to ping the Windows host because of the default firewall settings in Windows. You will modify the firewall rule later in this lab to allow the pings through the Windows Firewall. Press **CTRL-C** to stop the pings if necessary.

Step 2: Check for Windows updates.

a. In the **Windows Start Menu**, search for **check for updates**.

b. In the **Windows Update** window, you can review optional updates and the update history. Explore all the available options related to Windows Update and answer the following questions.

When was the last time the system checked for updates?

What are your current active hours? What will Windows do outside the active hours?

Step 3: Linux update and upgrade

a. In the **CSE-LABVM**, enter the command **apt-get** to see list of available commands. The **apt-get update** command should always be done before an upgrade.

b. Enter the command **sudo apt-get update** to resynchronize the package index files from their sources. Enter **password** as the password when prompted.

```
cisco@labvm:~$ sudo apt-get update
[sudo] password for cisco:
```

c. In the terminal, enter the command **sudo apt-get upgrade** to retrieve and upgrade the currently installed packages with new versions available. This command will not remove currently installed packages. If the newer version cannot be upgraded, no changes will be made to the packages.

Enter **password** as the password if prompted. Reply **y** when asked if you want to continue. This upgrade process can take a few minutes.

```
cisco@labvm:~$ sudo apt-get upgrade
[sudo] password for cisco:
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
<output omitted>
Need to get 479 MB of archives.
After this operation, 53.7 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Part 2: Windows Local Security Policy (Optional)

The Windows Local Security Policy of a system is a set of information about the security of your computer. In this part, you will configure the local password policy, account lockout settings, and the audit policy.

Note: Local Security Policy comes only with Windows Pro or Enterprise editions. If you have the Home edition, you can search the internet for tutorials on "How to Enable Local Security Policy (secpol.msc)." For

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example, the majorgeeks.com website has an excellent tutorial. If you are not allowed or would prefer not to change the Local Security Policy on you Windows host, read through this part and move on to the next part.

Step 1: Configure Local Password Policy in Windows

You have determined that the security policy for password is as follows:

A user must use a unique password for at least 2 changes of the password.

Passwords must be at least 8 characters.

Passwords must be changed every 90 days.

Can only change the password once a day.

A password must consist of the following four elements:

- At least one lower case alpha character.
- At least one upper case alpha character.
- At least one numerical character.
- At least one symbol character.

a. Navigate to the **Local Security Policy** by first searching for and opening the **Control Panel**.

b. Click **Administrative Tools** > Search for **Local Security Policy**.

c. Open **Local Security Policy**.

List a few of security policy settings:

d. Expand **Account Policies** and click **Password Policy**. Six policies are displayed in the right panel with their associated default security settings.

e. The first policy, **Enforce password history**, is used to set the number of unique passwords the user must enter before being allowed to reuse a password. Double-click **Enforce password history** to open the **Enforce password history Properties** window. Set the value to **2**.

f. Fill in the values you should set in **Local Security Policy** for the remaining **Password Policy** security settings.

Policy	Security Setting
Enforce password history	
Maximum password age	
Minimum password age	
Minimum password length	
Password must meet complexity requirements	
Store passwords using reversible encryption	

Note: The **Store passwords using reversible encryption** security setting should always be disabled. Storing passwords using reversible encryption is essentially the same as storing plaintext versions of the

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passwords. For this reason, this policy should never be enabled unless application requirements outweigh the need to protect password information.

g. Double-click each of the policies and set the values according to your entries in the table above.

Step 2: Test the password policy security settings.

Test your password policy security settings by attempting to change the password. Try a new password that does not meet the length or complexity requirements.

a. From the Start Menu, search for **Change your password**.

b. Click **Password**. Click **Change**.

c. Enter your current password. Click **Next** to continue.

d. Enter your new password twice and provide a password hint. Make sure your new password does not meet the length or complexity requirements that you configured in the previous step. Click **Next** to continue.

e. Click **Finish**. You should be presented with a message that your new password does not meet password policy requirements. Click **Close** to continue.

Step 3: Configure the Account Lockout Policy security settings.

a. Navigate back to the Local Security Policy window.

b. In the expanded **Account Policies** and click **Account Lockout Policy**. Three policies are displayed in the right panel with their associated default security settings.

c. Change the default settings to the following:

- A user must wait 10 minutes for the lookout counter to reset.
- Users are locked out of the computer after 5 attempts to enter the correct password.

How long should the user have to wait before attempting to log back in?

How many times is a user allowed to attempt to login before the account is locked?

Step 4: Configure the Audit Policy security settings.

a. Expand the **Local Policies** menu, and then click **Audit Policy**.

b. Double-click to open the properties window for **Audit account logon events**.

c. In the **Local Security Setting** tab, notice the check boxes for **Success** and **Failure**.

d. Click the **Explain** tab to learn about this security setting. Click **OK** to close the **Properties** window.

e. Continue to review each security setting. Click the **Explain** tab for each and read what it does.

Part 3: Configure Firewall Rules

Traffic travels in and out of devices using ports. The firewall controls the flow of the traffic. Think of the firewall as a security guard who controls the inbound and outbound traffic based on the firewall rules.

In this part, you will configure Windows Defender Firewall on a Windows.

Step 1: Investigate Windows Defender Firewall

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a. From the Start Menu, Search for and open **Windows Defender Firewall**. The default state for the Windows Firewall is On.

Note: If you are using a Windows PC that is administered by an organization, you may see the message, **For you security, some settings are managed by your system administrator.**

What are the benefits of Windows Firewall?

b. In the left panel of the window, click **Allow an app or feature through Windows Defender Firewal**l. In the **Allowed apps and features** window, programs and services that Windows Firewall is not blocking will be listed with a check mark.

Note: You can add applications to this list. This may be necessary if you have an application that requires outside communications but for some reason the Windows Firewall cannot perform the configuration automatically.

Creating too many exceptions in your Programs and Services file can have negative consequences.

Describe a negative consequence of having too many exceptions.

c. Click **Cancel** to exit the Allow apps window.

Step 2: Configure the Advanced Security features in Windows Firewall to allow echo requests.

Note: This step may not be allowed by your organization's security policy.

In this step, you will create an inbound rule that will allow echo request packets through the firewall.

a. In the left panel of the Windows Defender Firewall, click **Advanced settings**.

b. In the **Windows Defender Firewall with Advanced Security on Local Computer**, you can configure Inbound Rules, Outbound Rules, or Connection Security Rules. You can also click **Monitoring** to view the status of configured rules.

c. Click **Inbound Rules**, and then in the **Actions** pane, click **New Rule**.

d. In the **New Inbound Rule Wizard**, select **Custom**, and then click **Next** twice. You should now be at the **Protocol and Ports** step.

e. For **Protocol type**, select **ICMPv4**, and then click **Customize**.

f. In the **Customize ICMP Settings** window, select **Specific ICMP types**, select **Echo Request**, and then Click **OK**.

g. Click **Next** three times. You should now be at the **Profile** step.

h. Deselect **Public** so the Windows PC will not respond to an echo request in a public network location, such as an internet cafe. Click **Next** to continue.

i. Provide a name for the new inbound rule that gives a good description of the rule and click **Finish**. You should now see your rule at the top of the list of **Inbound Rules** in the **Windows Defender Firewall with Advanced Security** dialog window.

j. Now the rule has created and enabled. Verify that **CSE-LABVM** can ping the Windows host and receive replies.

Part 4: Install and Run Applications

In this part, you will install two new applications on **CSE-LABVM**: **chkrootkit** and **lynis**. The application **chkrootkit** will be downloaded from a software repository. However, we will add a new repository so we can install **lynis**, provided by CISOfy.

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Step 1: Install and run chkrootkit

The tool **chkrootkit** is used to check for signs of a rootkit on a local system. Rootkit is a type of malware that can remain hidden on your computer and can be used to cause significant damage to your device by hackers.

a. In a terminal, enter the command **sudo apt install chkrootkit**. Enter the password **password** when prompted.

```
cisco@labvm:~$ sudo apt install chkrootkit
[sudo] password for cisco:
```

b. Enter the command **sudo chkrootkit** to run a rootkit check.

```
cisco@labvm:~$ sudo chkrootkit
```

c. The output can be filtered to look for interested strings, such as worm. The command **chkrootkit** can be piped together with the command **grep** with the option **–i** to ignore the case distinction in the strings of interest.

```
cisco@labvm:~$ sudo chkrootkit | grep -i worm
Searching for LPD Worm files and dirs... nothing found
Searching for Ramen Worm files and dirs... nothing found
Searching for Adore Worm... nothing found
Searching for ShitC Worm... nothing found
Searching for Omega Worm... nothing found
Searching for Sadmind/IIS Worm... nothing found
Searching for TC2 Worm default files and dirs... nothing found
! cisco 32822 pts/0 grep --color=auto -i worm
```

Step 2: Install Lynis

lynis is security tool for systems running Unix-based OS, such as Linux and macOS. **lynis** will be used later in another activity to harden a Linux system. The application **Lynis** is maintained by CISOfy. In this step, we will add the software repository and install Lynis.

a. Copy and paste the following command into a terminal to import the key from the CISOfy keyserver. This key is required to verify the integrity of your download when you download **lynis**:

```
cisco@labvm:~$ sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-keys
013baa07180c50a7101097ef9de922f1c2fde6c4
Executing: /tmp/apt-key-gpghome.8C6X477onz/gpg.1.sh --keyserver keyserver.ubuntu.com -
-recv-keys 013baa07180c50a7101097ef9de922f1c2fde6c4
gpg: key FEBB7D1812576482: public key "CISOfy software signing <software@cisofy.com>"
imported
gpg: Total number processed: 1
gpg: imported: 1
```

b. Copy and paste the following command into a terminal to add the **lynis** repository maintained by CISOfy.

```
cisco@labvm:~$ echo "deb https://packages.cisofy.com/community/lynis/deb/
stable main" | sudo tee /etc/apt/sources.list.d/cisofy-lynis.list
deb https://packages.cisofy.com/community/lynis/deb/ stable main
```

c. Perform an update after adding a new repository. At the prompt, enter **sudo apt-get update**.

d. Use the command **apt install** to install Lynis.

```
cisco@labvm:~$ sudo apt install lynis
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

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```
The following NEW packages will be installed:
lynis
0 upgraded, 1 newly installed, 0 to remove and 17 not upgraded.
Need to get 0 B/262 kB of archives.
After this operation, 1,681 kB of additional disk space will be used.
Selecting previously unselected package lynis.
(Reading database ... 205787 files and directories currently installed.)
Preparing to unpack .../lynis_3.0.6-100_all.deb ...
Unpacking lynis (3.0.6-100) ...
Setting up lynis (3.0.6-100) ...
Processing triggers for man-db (2.9.1-1) ...
```

From the output, what is the installed Lynis version?

e. To verify the installed version, enter the command **lynis show version** at the terminal.

```
cisco@labvm:~$ lynis show version
3.0.6
```

f. If you want to determine the latest version provided by CISOfy, enter the following command at the terminal.

```
cisco@labvm:~$ sudo apt-cache policy lynis
lynis:
Installed: 3.0.6-100
Candidate: 3.0.6-100
Version table:
*** 3.0.6-100 500
500 https://packages.cisofy.com/community/lynis/deb stable/main amd64 Packages
500 https://packages.cisofy.com/community/lynis/deb stable/main i386 Packages
100 /var/lib/dpkg/status
2.6.2-1 500
500 http://archive.ubuntu.com/ubuntu focal/universe amd64 Packages
500 http://archive.ubuntu.com/ubuntu focal/universe i386 Packages
```

g. You can run **sudo apt-get update** and **sudo apt-get upgrade** again to ensure that you have all the latest updates from CISOfy.

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